NAVJOT SINGH CHAHAL

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Passionate Engineering graduate (June 2026) having 5+ years of professional experience in research & development of efficient and scalable autonomous systems.

EDUCATION

University of Pennsylvania, *Philadelphia, USA* *June 2026*

Master of Science *(Electrical Engineering at Department of Electrical Systems Engineering), GPA: 3.49/4.0*

Kurukshetra University, *Kurukshetra, India* *June 2019*

Bachelor of Technology *(Electronics and Communications Engineering), GPA: 4.0/4.0*

PROFESSIONAL EXPERIENCE

University of Pennsylvania,

Research Assistant at Figueroa GRASP Lab ***Jan 2026 – Present***

- Independent study on Proprioception sensory awareness in robots for low level control, aiding exteroception sensory like vision and active sensors. Building manufacturer agnostic data acquisition libraries in C++ and python and modelling the environmental awareness for Pybullet simulators, Franka robot arms. Inspired by human muscle system.
- Building inference characteristics on current, torque relationships designing model for auto tuning on any platform.

Research Assistant at Mod Lab ***Aug 2025 – Present***

- Developing string actuators for a modular robotic platform Abott.
- Working on defining scope of multi agent system for analysis of proprioceptive actuator data for task cognitive perception to reinforce the motor control or mutate physical agent configuration as independent Agents.

Teaching & Research Assistant ESE 5090 QHEL Lab ***Aug 2025 - Dec 2025***

- Developed an interactive Python notebook as an interactive initiative to perform experimental validations of quantum gates on actual devices from IBM cloud.
- Implemented various quantum algorithms like Grover's Algorithm, Shor's Algorithm, Quantum Fourier Transform, Bernstein-Vazirani Algorithm, Simon's Algorithm, Quantum cryptography.
- Engineering quantum control apparatus using Digital programmable function generator Moku-go for arbitrary function generation for Quantum spin control involving manipulating the intrinsic angular momentum (spin) of protons in Hydrogen atom to prepare, evolve, and measure quantum states.

Teaching Assistant ESE 5180 IoT Wireless, Security, & Scaling ***Aug 2025 – Dec 2025***

- IOT edge scaling with Nordic semiconductors to build systems CMake , low energy communications Lora , BLE , NFC firmware .
- Mentoring teams of 4 students in developing and securing guidance from external partners in academia and industry for developing embedded products and solutions

Lead Robotics Teacher SMP ***May 2025 – Aug 2025***

- Deployed optimization and ML base control techniques on the XRP pi, manipulations for the Robotic Arm Prehension and adding sensory cyber physical perception integrations. All these modules are custom built and developed for training 20 selected students from across US.
- Developed control infrastructure using embedded Linux (Jetson Nano, Raspberry Pi) for real-time manipulation tasks on a Castor Bot.

UHG Optum, ***Software Engineer - Fortune 5 CTO Acceleration Team, India*** ***March 2022 – Jul 2024***

- Scala 3 Play 2.8 back-end used functional programming paradigm and immutability to transform the large data streams using Akka Actors for managing multithreading to feed various ingesting systems.
- Built an Optum proprietary common versioning system Library packaged as ivy and maven for tool configurations with capabilities to Export and import and backup from AWS buckets and an extensible abstract class for supporting any database system.

HCL Technologies, ***Software Engineer, India*** ***Jul 2019 – March 2022***

- Developed SpringBoot-based embedded I/O control and NLP pipelines for industrial-grade document automation.
- Performed a POC for TWAIN-based web scanning using a third-party library and integrated I/O peripherals like a Printer/Scanner using Java, Azure.

- Integrated NLP based Human script recognition ML engine based on transformers to automate international cheque review systems of Deutsche Bank improving operational time and accuracy by 48 times.

PROJECTS

Heterogeneous Computing Pipeline for Stream Compression (C++ & HLS) SOC ***2025***

- Engineered a custom FPGA Kernel, CPU neon intrinsics and modified lookup table for hardware level synthesis.
- Implemented: Lempel-Ziv-Welch custom Kernel developed for (FPGA synthesis), Content defined chunking (Gear hash table) CPU Neon intrinsics accelerated SHA256 (CPU) A C++ OpenCL Kernel Host programming to communicate and control the Pipeline Kernel with Data flow parallelism.
- Achieving state of the art 135 Mbps per thread throughput only limited to be scaled by number of cores and kernel synthesis real estate you have for horizontal scaling.

Autonomous F1/10th Robot Racers Car Agent High Performance | University of Pennsylvania X-Lab ***2025***

- Interfaced Lidar, depth camera, PWM controllers like Vesc on Nvidia Jetson using ROS.
- Localization Navigation, Sensing, RL, sophisticated control software for autonomous robotic agents, MPC (Model Predictive Control), SLAM, Particle Filtering, Pure Pursuit (2nd Rank University Competition).
- Nerf Based 3D perception for localization and path planning poc completed using scene reconstruction.
- Building Kinematic models using cardan angles, curvature controllers and sensitive path planners, Kalman Filters, Markovian, Optimisation algorithms safety systems and multithreaded Nodes.
- Designing and building frameworks and tools for state-of-the-art mobility in vehicles. This includes middleware components such as scheduling for real-time systems, inter-process communication, and operating system abstractions.

Agentic AI Manager cross model communication and scaling | Scala | Light Bend Akka challenge ***2025***

- Built a Akka Actor system manager for multimodal scaleup, inter model communication and empowering the models with functional programming tools to achieve real world objectives.
- Built Agent selector, load balancing and Actor based asynchronous communication message protocol. I used Scala demonstrating the power of true functional programming in building modular frameworks for Agentic abilities and able to manage clusters of various capability ML models for building true cyber intelligent systems that can achieve meaningful automation in the world.

Risc V Microprocessor implemented on Lattice FPGA| University Of Pennsylvania ***2025***

- Built a RISC V ISA compliant microprocessor with multicycle, 5 stage pipelining, and a custom 8 stage Divider Module for optimizing DIV instructions, achieving 17.65 MHz Clock speed, with 4 stage AXIL4 Lite cache.

Low Light Image Enhancement using Transformer based neural network | University of Pennsylvania ***2024***

- Based on RetinexFormer, a transformer-based model with modifications in loss criterion Achieved a PSNR of 22.7 and live video enhancement at 5 FPS. Further accelerating the model using Quantization, pruning, and vertical acceleration on GPU (high-performance computing).

Deep Neural Network model compression and performance using CSR | University of Pennsylvania ***2024***

- Engineered custom sparse linear layer and implemented model pruning, quantization, and hard pruning techniques to compress large-scale deep neural networks.
- Achieved a 75% reduction in memory footprint of the VGG16 model (from 500+MB to 277MB) while preserving inference accuracy—enabling deployment on constrained edge platforms.
- Conducted comparative hardware acceleration benchmarking between Apple Metal Performance Shaders (MPS) and NVIDIA CUDA, focusing on throughput, latency, and model portability for heterogeneous compute architectures.

TECHNICAL SKILLS

- Languages: C, C++, CUDA, Python, Bash, Scala, Java, MATLAB, System Verilog
- Frameworks: ROS2, PyTorch, TensorFlow, Keras, TensorRT, OpenCV, OpenCL
- Tools: NVIDIA Jetson, Isaac Sim/Gym, Rviz, Docker, Git (Version Control), Linux, GDB, Vitis HLS
- Embedded/Robotics: LiDAR, Depth, IMU, Radar, MPC, SLAM, Localization, RTOS, Xilinx FPGA
- Concepts: Deep Neural Networks, Image Processing, Physics Simulation, Threads/Processes/Memory Synchronization, Optimization, Cloud Technologies (Azure, AWS, Docker).